

COMPANION SLIDES

Chapter 5

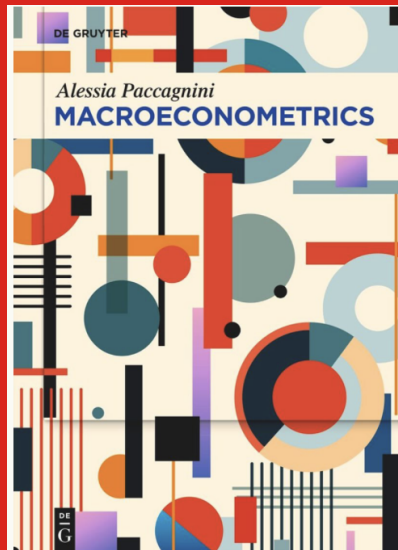
Bayesian Estimation

Priors, Posteriors and Bayesian VARs

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Data tell us something. Prior information tells us something.

Bayesian inference combines both.



- ▶ Explain Bayesian versus classical inference.
- ▶ Specify and interpret prior distributions.
- ▶ Derive posterior distributions in conjugate models.
- ▶ Understand Metropolis–Hastings and Gibbs sampling.
- ▶ Diagnose MCMC convergence.
- ▶ Estimate BVARs with Minnesota priors.
- ▶ Understand large BVAR shrinkage.
- ▶ Select hyperparameters hierarchically.
- ▶ Handle extreme observations with pandemic priors.
- ▶ Compare models and conduct structural Bayesian analysis.

Posterior distribution

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{p(y)} \propto p(y | \theta)p(\theta).$$

Prior

What we know before seeing the sample.

Likelihood

What the sample says about the parameters.

Posterior

Updated uncertainty after combining both.

	Classical	Bayesian
Parameters	fixed, unknown	random variables
Uncertainty	sampling distribution	posterior distribution
Prior information	not explicit	incorporated through $p(\theta)$
Intervals	confidence intervals	credible intervals
Model comparison	tests / IC	marginal likelihood / Bayes factors

Key takeaway

Bayesian probability statements are directly about uncertain parameters conditional on the observed data and prior.

- ▶ Macroeconomic samples are often short relative to model dimension.
- ▶ VARs proliferate parameters rapidly as variables and lags increase.
- ▶ Priors provide **regularization**: implausible dynamics can be shrunk toward parsimonious benchmarks.
- ▶ Posterior distributions propagate parameter uncertainty naturally into forecasts and IRFs.
- ▶ Prior structure can encode economically meaningful beliefs while remaining transparent and testable through sensitivity analysis.

Sources of prior information

- ▶ economic theory;
- ▶ previous empirical evidence;
- ▶ persistence benchmarks;
- ▶ regularization / shrinkage goals.

A good applied question

Is the prior expressing substantive knowledge, stabilizing estimation, or both?

Sensitivity analysis should show which conclusions depend materially on the prior.

Core idea

A conjugate prior is chosen so that the posterior belongs to the same family as the prior.

- ▶ Linear Gaussian models admit convenient Normal / Inverse-Wishart structures.
- ▶ Closed-form posteriors can make simulation straightforward.
- ▶ Conjugacy is especially useful for BVARs because posterior draws can often be generated directly.

Key takeaway

Not every Bayesian model requires a long MCMC chain.

Metropolis–Hastings

Propose a candidate parameter draw and accept or reject it according to a probability based on the posterior ratio.

Gibbs sampling

Draw sequentially from conditional posterior distributions when these are easier to sample than the joint posterior.

The objective is to approximate the posterior distribution with simulated draws.

- ▶ Inspect trace plots: does the chain explore the posterior rather than stick locally?
- ▶ Check autocorrelation and effective sample size.
- ▶ Compare multiple chains / starting values when useful.
- ▶ Separate burn-in from retained draws when the algorithm requires it.
- ▶ Poor tuning can create slow mixing and misleading posterior summaries.

Rule

Never treat a large number of iterations as proof of convergence.

The parameter proliferation problem

A VAR with K variables and p lags has roughly K^2p slope coefficients before constants and deterministic terms.

- ▶ Adding information can improve forecasting and structural analysis.
- ▶ But unrestricted estimation becomes noisy as dimension grows.
- ▶ Bayesian shrinkage allows a large information set while disciplining weakly identified coefficients.

Prior mean

$$E[A_{ij}^{(l)}] = \delta_i \mathbf{1}\{i = j, l = 1\}.$$

- ▶ Own first lags are centered near a persistent / random-walk benchmark when appropriate.
- ▶ Cross-variable coefficients are centered at zero.
- ▶ Higher lags receive progressively tighter shrinkage.
- ▶ Overall tightness controls how strongly the model is pulled toward the prior.

λ_1

Overall tightness. Smaller values mean stronger shrinkage.

λ_2

Relative shrinkage on cross-variable lags.

λ_3

Lag decay: distant lags are increasingly penalized.

Key takeaway

Hyperparameters determine how much complexity the data are allowed to support.

- ▶ Minnesota-style beliefs can be implemented by augmenting the sample with artificial observations.
- ▶ Dummy observations make prior beliefs look like additional data restrictions.
- ▶ The same framework can incorporate beliefs about persistence, sums of coefficients and long-run behavior.
- ▶ This representation is computationally convenient in conjugate BVARs.

The information problem

- ▶ Macroeconomic decisions use many indicators.
- ▶ Small VARs may omit relevant information.
- ▶ Large systems can contain dozens or hundreds of variables.

The shrinkage solution

- ▶ allow many predictors;
- ▶ shrink weak coefficients strongly;
- ▶ calibrate tightness to model dimension;
- ▶ retain useful information without unrestricted overfitting.

Instead of fixing shrinkage by hand

Treat hyperparameters as unknown and let the data inform them through the marginal likelihood.

$$p(y) = \int p(y | \theta) p(\theta) d\theta.$$

- ▶ The marginal likelihood rewards fit but penalizes unnecessary flexibility.
- ▶ Hierarchical selection reduces dependence on arbitrary prior tightness choices.
- ▶ It adapts shrinkage to the information content of the sample.

The COVID-19 problem

Extreme observations can distort estimated persistence and overwhelm a conventional VAR likelihood.

- ▶ Pandemic priors introduce time-specific components to absorb exceptional observations.
- ▶ The objective is to preserve underlying pre-pandemic dynamics while retaining information from the crisis period.
- ▶ Alternative approaches include variance scaling and stochastic volatility.

- ▶ The identification problem from Chapter 4 remains: Bayesian estimation does not automatically identify structural shocks.
- ▶ Cholesky, sign restrictions, narrative restrictions, external instruments and heteroskedasticity can all be embedded in Bayesian analysis.
- ▶ Sign restrictions are especially natural: draw reduced-form parameters and rotations, retain draws satisfying the restrictions.

Key takeaway

Bayesian estimation handles parameter uncertainty; structural identification still requires identifying information.

Marginal likelihood

Integrates the likelihood over the prior distribution and forms the basis for Bayesian model comparison.

Bayes factors

Compare the relative evidence for two models using the ratio of their marginal likelihoods.

Information criteria can also be reported, but they answer a different approximation problem than fully Bayesian model probabilities.

- ▶ Quarterly US data, 1970Q1–2007Q4.
- ▶ GDP growth, GDP-deflator inflation, Federal Funds rate.
- ▶ VAR(4).
- ▶ Cholesky ordering: GDP growth $\rightarrow$ inflation $\rightarrow$ Fed Funds.
- ▶ Minnesota prior with baseline $\lambda_1 = 0.2$.

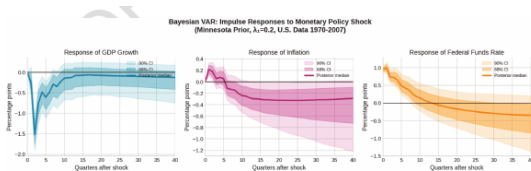


Figure E.3 • Bayesian VAR: Impulse Responses to Monetary Policy Shock

Conjugate Normal–Inverse–Wishart structure

The empirical example draws directly from the posterior: first $\Sigma^{(j)}$, then the VAR coefficients conditional on $\Sigma^{(j)}$.

- ▶ 2,000 independent posterior draws are used in the book example.
- ▶ For each draw, compute Cholesky-identified impulse responses.
- ▶ Summarize the resulting IRF distribution with posterior medians and credible intervals.

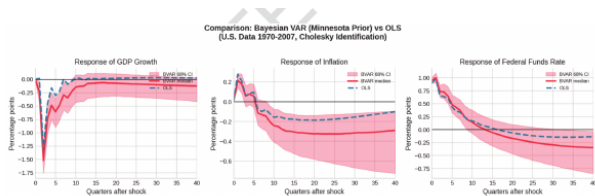


Figure 5.4: Comparison: Bayesian VAR vs OLS Estimation.

Notes: VAR posterior median (solid red) with 50% credible interval (shaded) compared to OLS point estimate (blue dashed line).

What changes?

Shrinkage pulls coefficients toward more parsimonious dynamics when the likelihood is weak, while persistent and well-estimated relationships can remain close to OLS.

The posterior distribution also provides a direct representation of parameter uncertainty.

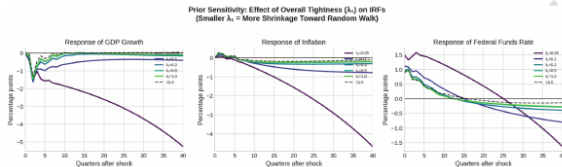


Figure 5.5: Prior Sensitivity: Effect of Overall Tightness (λ_1).

Notes: Impulse responses to monetary policy shock for different values of the overall tightness parameter.

The book varies λ_1 from strong to weak shrinkage.

- ▶ Smaller λ_1 : tighter prior.
- ▶ Larger λ_1 : likelihood has more influence.
- ▶ As the prior becomes diffuse, BVAR estimates move toward OLS.

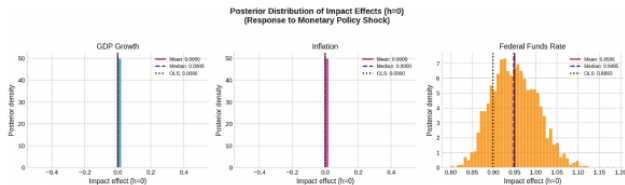


Figure 5.6: Posterior Distributions of Impact Effects.

Notes: Histograms show the posterior distribution of impact responses ($h = 0$) to a monetary policy shock.

Posterior distributions make uncertainty visible rather than reducing inference to a single coefficient estimate.

Interpretation

A credible interval is a probability statement about the parameter or response conditional on the model, data and prior.

A useful warning from the empirical example

Bayesian shrinkage can stabilize estimation, but it cannot repair an information set that omits variables relevant for identifying the policy shock.

- ▶ The inflation response can still display the price puzzle.
- ▶ The chapter links this to informational insufficiency in small-scale VARs.
- ▶ Large BVARs offer one route to richer information sets while controlling dimensionality through shrinkage.

1. Define the model and transformation of the data.
2. Choose a prior and explain its economic/statistical role.
3. Set or estimate hyperparameters.
4. Obtain posterior draws analytically or by MCMC.
5. Diagnose simulation quality when MCMC is used.
6. Compute forecasts, IRFs or other posterior objects.
7. Report credible intervals and prior sensitivity.
8. Compare plausible models / priors.

1. Bayesian inference combines prior information with the likelihood.
2. Shrinkage makes parameter-rich VARs estimable and often more stable.
3. The Minnesota prior is a transparent benchmark for BVAR regularization.
4. Hierarchical methods let the data inform shrinkage intensity.
5. Posterior uncertainty and prior sensitivity should be central parts of applied Bayesian reporting.

- ▶ Reestimate a BVAR for several values of λ_1 and compare IRFs.
- ▶ Compare OLS confidence intervals with Bayesian credible intervals.
- ▶ Expand the three-variable monetary VAR into a larger information set.
- ▶ Compare fixed versus hierarchical hyperparameters.
- ▶ Add pandemic observations and test alternative treatments of extreme data.
- ▶ Implement sign restrictions using posterior draws.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 5. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add derivations of posterior distributions.
- ▶ Add your preferred BVAR software and coding demonstrations.
- ▶ Add alternative priors, datasets and forecasting applications.
- ▶ Add exercises on MCMC diagnostics and prior sensitivity.

Make them yours.

DSGE Models

From reduced-form dynamics to structural economic models

Next question

How can economic theory, equilibrium conditions and Bayesian estimation be combined in a structural model of the macroeconomy?